API Fundamentals

API (Application Programming Interface) is a software intermediary that enables two applications to communicate with each other. It receives requests from a client, forwards them to the server, and returns the server's response.

Example: When a student logs into an LMS, the mobile app sends the username and password to the server through an API. The server verifies the credentials and sends back the login result.

API Flow

Client (LMS App)

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API

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Server

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Database

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Response

Frontend vs Backend

Frontend	Backend
The part of the application that users interact with.(buttons, checkboxes, forms)	The part of the application that processes requests and manages business logic behind the scenes.
Visible to users	Runs behind the scenes
Technologies: HTML, CSS, JavaScript	Technologies: Java, Python, .NET, Node.js
Displays data and collects user input	Processes data and returns responses
Sends API requests to the server	Receives API requests, processes them, and sends API responses
LMS Example: Displays the Login page, allows the student to enter a username and password, and sends the login request when the Login button is clicked.	**LMS Example:** Receives the login request, validates the student's credentials, checks the database, generates a login token if successful, and sends a success or failure response back to the frontend.

Client vs Server

Client	Server
The application used by the user to send requests to the server.	The system that receives requests, processes them, and returns responses.
Runs on the user's device (browser, laptop, or mobile app).	Runs on Remote server or cloud infrastructure.
Displays the user interface (UI), accepts user input, and sends API requests.	Executes business logic, validates requests, interacts with the database, and returns API responses.
Technologies: HTML, CSS, JavaScript, React, Angular, Flutter.	Technologies: Java, Python, .NET, Node.js, Spring Boot, Express.js.
LMS Example (Login): The student opens the LMS website, enters their username and password, and clicks Login. The client sends a POST /login API request to the server.	**LMS Example (Login):** The server receives the login request, validates the credentials against the database, generates a login token, and returns a success or failure response.
LMS Example (View Profile): The student clicks My Profile. The client sends a GET /api/student/101 request with the Authorization token in the request header.	**LMS Example (View Profile):** The server validates the token, retrieves the student's details from the database, and returns the profile information as a JSON response.

Communication flow

Student

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LMS Website (Client)

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POST /login

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LMS Server

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Database

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Login Successful

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Dashboard Displayed

API Testing

API Testing verifies that APIs function correctly without testing the user interface.

It checks:

- Request validation
- Response validation
- Status codes

- Authentication
- Data accuracy
- Performance
- Error handling
- Response Time